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MATH 247A

Homework 1
Due: Oct 09, 2026

Problem 11

If $1 < p < \infty$, determine the cases for which Minkowski's inequality

$$\left\| \sum_i f_i \right\|_p \leq \sum_i \|f_i\|_p$$

holds with equality. What changes when $p = 1$ or $p = \infty$?

Solution.

$$\begin{aligned}
\left\| \sum_{i \in I} f_i \right\|_p^p &= \int_X \left| \sum_{i \in I} f_i(x) \right|^p \mathrm{d}\mu \\
&= \int_X \left| \sum_{i \in I} f_i(x) \right|^{p-1} \left| \sum_{i \in I} f_i(x) \right| \mathrm{d}\mu \\
&\leq \int_X \left| \sum_{x \in I} f_i(x) \right|^{p-1} \left(\sum_{i \in I} |f_i(x)| \right) \mathrm{d}\mu
\end{aligned}$$

Note that the last inequality, arises due to the triangle inequality of complex numbers:

$$\left| \sum_{i \in I} f_i(x) \right| \leq \sum_{i \in I} |f_i(x)|.$$

Equality in the triangle inequality only holds if $f_i(x) = \lambda_i(x) f(x)$ for $\lambda_i(x) \in [0, \infty)$ and $f : X \to S^1$. Thus if $f_i = \lambda_i f$ almost everywhere, where $\lambda_i(x) \in [0, \infty)$ and $f : X \to S^1$ equality holds. So suppose this is the case.

$$\begin{aligned}
\left\|\sum_{i\in I} f_i\right\|_p^p &= \int_X \left(\left|\sum_{i\in I} f_i(x)\right|^{p-1} \sum_{i\in I} |f_i(x)|\right) \mathrm{d}\mu \\
&= \int_X \sum_{i\in I} |f_i(x)| \left|\sum_{j\in I} f_j(x)\right|^{p-1} \mathrm{d}\mu \\
&= \sum_{i\in I} \int_X |f_i(x)| \left|\sum_{j\in I} f_j(x)\right|^{p-1} \mathrm{d}\mu \qquad \text{(Tonell's Theorem)} \\
&= \sum_{i\in I} \int_X |f_i(x)| \left|\left(\sum_{j\in I} f_j(x)\right)^{p-1}\right| \mathrm{d}\mu \\
&= \sum_{i\in I} \left\| f_i(x) \left(\sum_{j\in I} f_j(x)\right)^{p-1} \right\|_1 \\
&\leq \sum_{i\in I} \|f_i(x)\|_p \left\|\sum_{j\in I} f_j(x)\right\|_{\frac{p}{p-1}}
\end{aligned}$$

The final inequality comes from Holder.

Problem 12

Show that Hölder's inequality

$$\|fg\|_r \leq \|f\|_p \|g\|_q, \quad \frac{1}{p} + \frac{1}{q} = \frac{1}{r}$$

is equivalent to the log-convexity of L^p norms: for $0 < p < q \leq \infty$, $0 < \theta < 1$, and $\frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}$,

$$\|f\|_r \leq \|f\|_p^{1-\theta} \|f\|_q^{\theta}.$$

Solution. $\Rightarrow$: Suppose Holder's inequality is true for all choices of parameters. Suppose

$$\frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}.$$

Let

$$\alpha = \frac{p}{1-\theta}$$
$$\beta = \frac{q}{\theta}.$$

Thus by Holder's inequality we have

$$\begin{aligned}\|f\|_r &= \left\|f^{1-\theta} f^{\theta}\right\|_r \\ &\leq \left\|f^{1-\theta}\right\|_{\alpha} \left\|f^{\theta}\right\|_{\beta} \\ &= \left(\int_X |f(x)|^{(1-\theta)\alpha}\,\mathrm{d}\mu\right)^{\frac{1}{\alpha}} \left(\int_X |f(x)|^{\theta\beta}\,\mathrm{d}\mu\right)^{\frac{1}{\beta}} \\ &= \left(\int_X |f(x)|^{p}\,\mathrm{d}\mu\right)^{\frac{1-\theta}{p}} \left(\int_X |f(x)|^{q}\,\mathrm{d}\mu\right)^{\frac{\theta}{q}} = \|f\|_p^{1-\theta}\|f\|_q^{\theta}.\end{aligned}$$

$\Longleftarrow$: Suppose the log-convexity of L^p is true. Let

$$\frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

Without loss of generality $f \in L^p(X, d\mu)$ and $g \in L^q(X, d\mu)$ (otherwise statement is trivially true). We will first show that it is sufficient to prove for (X, μ) where $\mu(X) < \infty$. Suppose Holder's inequality is true for such a X. Note that $\operatorname{supp}(f)$, $\operatorname{supp}(fg)$, and $\operatorname{supp}(g)$ are σ-finite, or a union of $A_n \subseteq X$ of finite measure where $A_n \subseteq A_{n+1}$. Let

$$\operatorname{supp}(f), \operatorname{supp}(g), \operatorname{supp}(fg) \subseteq \bigcup_{n\in\mathbb{N}} A_n.$$

Note because $\mu(A_n) < \infty$, we have that

$$\|fg\|_{L^r\left(A_n, d\mu|_{A_n}\right)} \leq \|f\|_{L^p\left(A_n, d\mu\,|_{A_n}\right)} \|g\|_{L^q\left(A_n, d\mu\,|_{A_n}\right)}$$

for all $n \in \mathbb{N}$. Let us examine each side:

$$\begin{aligned}\|fg\|_{L^r\left(A_n, d\mu|_{A_n}\right)} &= \left(\int_{A_n} |f(x)g(x)|^r\,\mathrm{d}\mu|_{A_n}\right)^{\frac{1}{r}} \\ &= \left(\int_X |f(x)g(x)|^r 1_{A_n}\,\mathrm{d}\mu\right)^{\frac{1}{r}}.\end{aligned}$$

By monotone convergence theorem and the fact that $x \to x^{\frac{1}{p}}$ is continuous:

$$\begin{aligned}\lim_{n\to\infty} \|fg\|_{L^r\left(A_n, \mathrm{d}\mu|_{A_n}\right)} &= \left(\int_X |f(x)g(x)|^r 1_{\cup_n A_n}\,\mathrm{d}\mu\right)^{\frac{1}{r}} \\ &= \left(\int_X |f(x)g(x)|^r\,\mathrm{d}\mu\right)^{\frac{1}{r}} \quad \left(\text{using} \operatorname{supp}(fg) \subseteq \bigcup_n A_n\right) \\ &= \|fg\|_{L^r(X,\,\mathrm{d}\mu)}.\end{aligned}$$

By a similar argument, we see that

$$\|f\|_{L^p\left(A_n, d\mu\,|_{A_n}\right)}\|g\|_{L^q\left(A_n, d\mu\,|_{A_n}\right)} \underset{n\to\infty}{\to} \|f\|_{L^p(X,d\mu)}\|g\|_{L^q(X,d\mu)}.$$

Thus we have Holder's inequality:

$$\|fg\|_r \le \|f\|_p\|g\|_q.$$

Thus it suffices to prove Holder's inequality for the case of X where $\mu(X) < \infty$. Furthermore, we can make the reduction that f, g are everywhere nonvanishing because $\text{supp}(f), \text{supp}(g) \subseteq X$ and thus must also be finite measure, meaning we can restrict to them and nothing would change. Without loss of generality $f = |f|$ and $g = |g|$. We will now reduce to the case of simple functions. Suppose Holder's inequality is true for simple functions. f, g are approximated by sequence of simple functions φ_n and ψ_n respectively each also lying in p, q (respectively) and $\varphi_n \le \varphi_{n+1} \le f$ and $\psi_n \le \psi_{n+1} \le g$. By monotone convergence, Holder's inequality holds for f, g. Thus the problem reduces to proving Holder's inequality for f, g simple functions. Note it is sufficient to prove that if $\|f\|_p = 1$ and $\|g\|_q = 1$ then $\|fg\|_r \le 1$. Furthermore, suppose we prove it for $\alpha, \beta \in (0, \infty)$ where

$$1 = \frac{1}{\alpha} + \frac{1}{\beta},$$

and $\|\psi\|_\alpha = 1$ and $\|\varphi\|_\beta = 1$ implies that $\|\varphi\psi\|_1 \le 1$. Let $\alpha = \frac{p}{r}$ and $\beta = \frac{q}{r}$. Let

$$\psi = f^r$$
$$\varphi = g^r.$$

$$\begin{aligned}
\|f^r\|_\alpha &= \left(\int_X |f^r|^{\frac{p}{r}}\,\mathrm{d}\mu\right)^{\frac{r}{p}} \\
&= \|f\|_p^r = 1 \\
\|g^r\|_\beta &= \left(\int_X |g^r|^{\frac{q}{r}}\,\mathrm{d}\mu\right)^{\frac{r}{q}} \\
&= \|g\|_q^r = 1 \\
\Rightarrow 1 &\ge \|(fg)^r\|_1 \\
&= \int_X (|fg|^r)\,\mathrm{d}\mu \\
&= \|fg\|_r^r \Rightarrow \|fg\|_r \le 1
\end{aligned}$$

Thus the last reduction tells us it is sufficient to prove for r = 1. Consider

$$\mathrm{d}\nu = |g|^q\,\mathrm{d}\mu$$
$$h(x) = \frac{1}{|g(x)|^{\frac{q}{p}}} f(x).$$

Thus, we can match some norms

$$\begin{aligned}\|h\|_{L^1(X,\mathrm{d}\nu)} &= \int_X |h(x)||g(x)|^q \,\mathrm{d}\mu \\ &= \int_X |f(x)||g(x)|^{q-\frac{q}{p}} \,\mathrm{d}\mu \\ &= \int_X |f(x)||g(x)| \,\mathrm{d}\mu \\ &= \|fg\|_{L^1(X,\mathrm{d}\mu)}\end{aligned}$$

$$\begin{aligned}\|h\|_{L^p(X,\mathrm{d}\nu)} &= \left(\int_X \|f\|^p \|g\|^{q-q} \,\mathrm{d}\mu\right)^{\frac{1}{p}} \\ &= \|f\|_p = 1.\end{aligned}$$

By log convexity, $\forall q'$ where $0 < q' < 1 < p$ and $\theta \in (0,1)$ where

$$1 = \frac{1-\theta}{q'} + \frac{\theta}{p}$$

we have that

$$\begin{aligned}\|fg\|_{L^1(X,\mathrm{d}\mu)} = \|h\|_{L^1(X,\mathrm{d}\nu)} &\le \|h\|^{\theta}_{L^p(X,\mathrm{d}\nu)} \|h\|^{1-\theta}_{L^{q'}(X,\mathrm{d}\nu)} \\ &= \|h\|^{1-\theta}_{L^{q'}(X,\mathrm{d}\nu)}\end{aligned}$$

Note that as $q' \to 0^+$ we have that $\theta \to 1$ because

$$1 - \theta = q'\left(1 - \frac{\theta}{p}\right).$$

Since $q' \to 0$ and $h > 0$ positive simple function, $h^{q'} \to 1$ everywhere and hence $\int_X h^{q'} \,\mathrm{d}\nu \to \int_X \mathrm{d}\nu = \nu(X)$. Thus

$$\|h\|_{L^1(X,\mathrm{d}\nu)} \le \lim_{q' \to 0} \|h\|^{1-\theta}_{L^{q'}(X,\mathrm{d}\nu)} = \nu(X) = \|g\|_{L^q(X,\mathrm{d}\mu)} = 1.$$

Thus putting all this together, we see that

$$\|fg\|_{L^1(X,\mathrm{d}\mu)} \le 1.$$

Problem 13

Differentiate $\log \|f\|_p$ twice with respect to $\alpha := \frac{1}{p}$ and show that this is non-negative (take f to be a non-zero simple function with finite measure support, to avoid technicalities).

Solution. Let f be a non-zero simple function with finite measure support (we can just assume $\mu(X) < \infty$). Thus $\|f\|_p \neq 0$ for all $p \in (0,\infty)$. Note that since f simple and finite measure support, $\|f\|_p < \infty$ for all $p \in (0,\infty)$. Furthermore, since f simple,

$$f = \sum_{k=1}^{n} c_k 1_{E_k}$$

for $E_k \subset X$ and $c_k \in \mathbb{R}$. Let $C = \max(c_1, ..., c_n)$. Let

$$I(\alpha) := \int_X |f(x)|^{\frac{1}{\alpha}} \, \mathrm{d}x$$

Problem 18

On a measure space consisting of N points (with counting measure), and $0 < p \leq q \leq \infty$, one has

$$\|f\|_{\ell^q} \leq \|f\|_{\ell^p} \leq N^{\frac{1}{p}-\frac{1}{q}} \|f\|_{\ell^q}.$$

When does equality occur for either of these two inequalities? Note how the example that attains the lower bound is in many ways the "opposite extreme" to the example which attains the upper bound.

Solution.

Problem 19

Establish the (vector-valued) bound

$$\left\| \left(\sum_n |f_n|^q \right)^{\frac{1}{q}} \right\|_{L^p} \leq \left(\sum_n \|f_n\|_{L^p}^q \right)^{\frac{1}{q}}$$

for any measurable functions f_n and any $0 < q < p \leq \infty$.

Solution.